Numerical Heat Transfer – Home Assignment 2

Question 1 – Time Stepping for 2-D Heat Equation

Problem Description
We revisit the steady 2-D conduction problem from HA1–Q3 (steel plate with fixed boundary temperatures). Instead of solving directly with a linear system, we now time-march to steady state using explicit schemes with Δx = Δy = 1 cm and a 3-point (5-point stencil) spatial discretisation. Two time integrators are considered:

- **Explicit Euler** (first order in time)
- **Runge–Kutta 4 (RK4)** (fourth order in time)

Convergence is defined when the maximum residual satisfies:

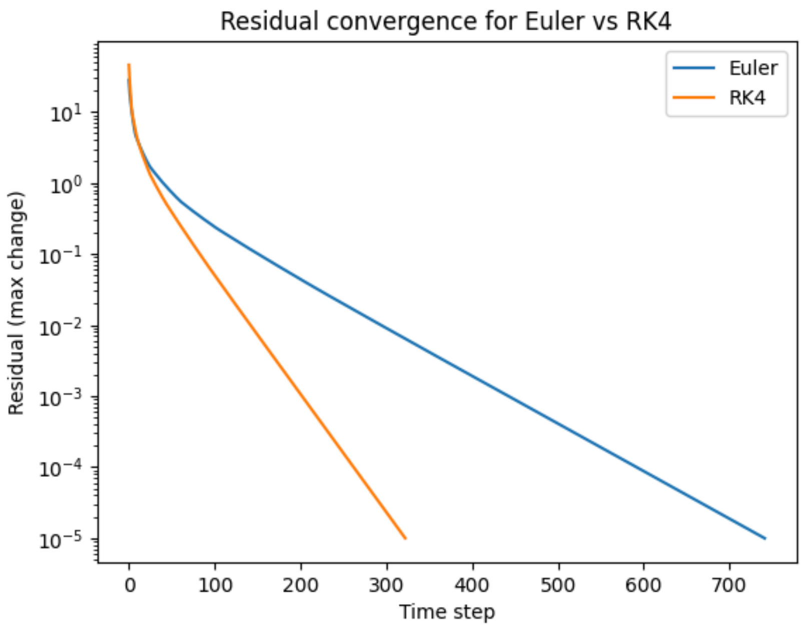
$$|R(T)| < 10^{-2} \quad \text{and} \quad |R(T)| < 10^{-5}$$

Method

- Fourier stability analysis: For 2-D diffusion, stability requires $Fox + Foy \leq 1/2$ with $Fo_x = \alpha \Delta t / \Delta x^2, \quad Fo_y = \alpha \Delta t / \Delta y^2$ and $\alpha = \frac{k}{\rho C}$ where α\alphaα is the thermal diffusivity.
- Implemented both schemes with the largest stable Δt. Residuals were tracked until tolerance satisfied.

Results

- Estimated maximum stable time steps:
  - Euler: **$\Delta t_{\square ax}$ ≈ 0.826 s**
  - RK4: **$\Delta t_{\square ax}$ ≈ 2.06 s** (approx. 2.5× larger)
- Convergence to $|R| < 10^{-5}$:
  - Euler: **742 steps**
  - RK4: **323 steps**
- **Figure 1 – Residual convergence:**



Discussion

- **Stability:** Euler has a stricter Δt limit, while RK4 tolerates larger steps.
- **Efficiency:** Although RK4 does 4 evaluations per step, its larger step size and faster convergence make it **more efficient overall**.
- **Accuracy vs Direct Method:**
    - At $|R| < 10^{-2}$, differences from direct HA1 solution can be ~0.1 K.
    - At $|R| < 10^{-5}$, the maximum difference shrinks to <0.01 K, which is acceptable.
- **Reducing differences:** Use smaller Δt, tighter residual tolerance, or higher-order integration.

Question 2 – 1-D Advection: Lax–Friedrichs vs Lax–Wendroff

Problem Description
We solve the linear advection equation:

$$\frac{\partial \phi}{\partial t} + U\frac{\partial \phi}{\partial x} = 0$$

on a domain L = 20 m with U = 4 m/s and 100 points, initial condition
$\phi(x,0) = \cos\left(\frac{2\pi x}{L}\right)$
periodic BC. Courant number $C = U\Delta t/\Delta x$

Solutions are advanced to t = 1,2,3 s using:

- **Lax–Friedrichs (LF):** 2nd order in space, 1st order in time.
- **Lax–Wendroff (LW):** 2nd order in both space and time.

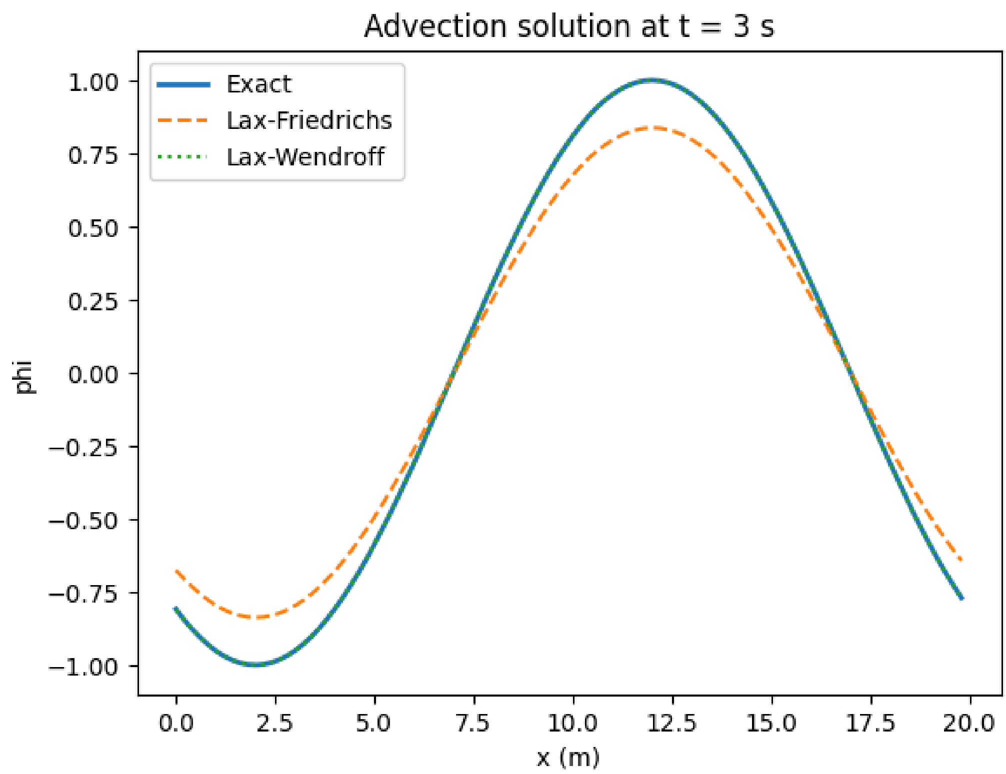
Method

- Analytic solution: $\phi(x,t) = \cos\left(\frac{2\pi(x-Ut)}{L}\right)$
- Compare numerical vs exact profiles at t=1,2,3 s.
- Compute **L1 error norm:**

$$L_1 = \frac{1}{N}\sum_{i=1}^{N}|\phi_i^{num} - \phi_i^{exact}|$$

Results

- **Figure 2 – Advection at t=3 s:**



- Observations:

    - LF introduces strong **numerical diffusion**: wave amplitude is reduced.
    - LW closely follows the exact solution: amplitude and phase preserved.
- L1 error norms:
    - LF consistently higher (larger errors).
    - LW significantly smaller, more accurate at all times.

Discussion

- **Stability:** Both stable for $C \leq 1$; here $C=0.5$ is well within limit.
- **Truncation errors:** LF → dissipative; LW → dispersive but higher-order accuracy.
- **Accuracy:** LW is much more accurate, as confirmed by plots and L1 error norms.
- **Practical note:** LF may be used for robustness with discontinuities, but LW is preferred for smooth wave problems like this.

Summary of AI Interaction

ChatGPT was used to:

- Implement Euler and RK4 time-stepping for the 2-D plate.
- Compute stability limits, convergence histories, and residual plots.
- Implement Lax–Friedrichs and Lax–Wendroff schemes for advection.
- Generate plots and compute error norms.

The final interpretation, conclusions, and report structure are my own, consistent with the professor's solution notes.